A Quantitative Theory of the Intangible Revolution

S. Dyrda[1], G. Hong[2], A. Sajid[1], J. Steinberg[1] | [1]University of Toronto,[2] Michigan State University

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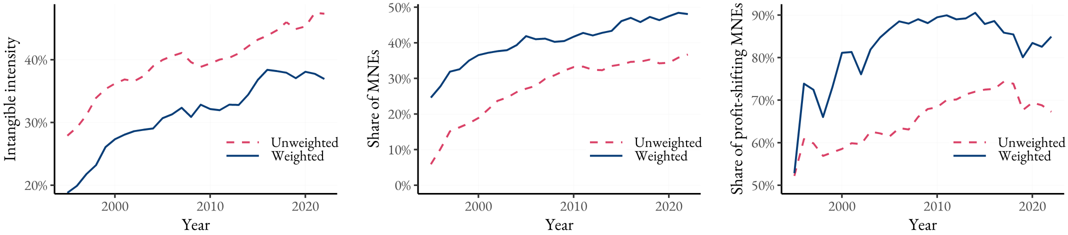
The issue and why do we care?

Introduction

- **Secular Trends**—Since the 1990s, **intangible intensity, multinational production, and profit-shifting** have surged dramatically. data



(a) Intangible Intensity *(b)* Share of MNEs *(c)* Share of Profit-shifting MNEs

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Introduction

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- **Why It Matters**—These shifts redefine how firms create value and where profits are booked:
  - → Intangible capital has become central to production (Crouzet & Eberly, 2021)
  - → R&D and production have decoupled geographically (Bernard & Fort, 2017)
  - → Profits are increasingly shifted to low-tax jurisdictions (Tørsløv et al., 2023)

- **Explaining the "intangible revolution"** requires a unified framework connecting intangibles, globalization, and profit shifting — which is exactly **our objective**.

What we do - Roadmap

Introduction

1. **Empirical Linkage**

   Rising intangible intensity, MNE expansion, and profit-shifting **are linked empirically** using consistent firm-level data.

2. **Structural Linkage**

   Those same trends **are linked structurally** in a stylized analytical model, with nonrival intangibles, FDI decisions, and profit-shifting.

3. **Quantitative Synthesis** (in progress)

   The quantitative model integrates these elements, illustrating how **intangible prices and declining FDI barriers** drive the observed co-movements.

Key takeaways - What we find

Introduction

1. **Empirical evidence**
   → MNEs and profit shifters have substantially **higher intangible intensity** in the cross-section, and when firms become MNEs or start shifting profits, their intangible intensity rises further.
   → High-TFP, high-intangible firms are the ones that **expand abroad**.

2. **Theory**
   → Falling costs of intangibles, expansion, and profit shifting **reinforce each other** (complementarity), and their effects are amplified for **more productive firms**.

3. **Quantitative Model** (in progress)
   → Produces **selection patterns** consistent with the data.
   → Generates **non-negligible complementarities** between reductions in the marginal cost of intangibles and expansion costs.

Literature review

Introduction

- **Intangible capital and growth**: McGrattan and Prescott (2010), Chappell and Jaffe (2018), Crouzet and Eberly (2021), Corrado, Haskel, Jona-Lasinio and Iommi (2022), De Ridder (2024), Li (2025)

- **Gains from globalization**: Ramondo and Rodríguez-Clare (2013), Tintlenot (2017), Arkolakis, Ramondo, Rodríguez-Clare and Yeaple (2018), Burstein and Monge-Naranjo (2024), Gumpert, Manova, Rujan and Schnitzer (2025)

- **Tax base erasion through transfer pricing of intangibles**: Dischinger and Riedel (2011), Accoto, Federico and Oddo (2021), Schwab and Todtenhaupt (2021), Guvenen, Mataloni, Rassier and Ruhl (2022), Santacreu (2023), Dyrda, Hong and Steinberg (2024)

Empirical Evidence

Data

Empirics

- Firm balance-sheet data from **Compustat North America**
  - → Consolidated financial statement
  - → Information on sales, tangible and intangible capital, investment, foreign income, and tax liabilities
- **Intangible capital**
  - → Balance-sheet measure includes mostly externally-acquired intangibles
  - → Internally-generated intangible capital measured using the perpetual inventory method from R&D expenditures (Peters and Taylor, 2017)
- **Multinational activity**
  - → Extensive margin: firms that have foreign subsidiaries reported in Exhibit 21 of 10-K forms
  - → Intensive margin: the number of foreign countries and the share of foreign income
- **Profit-shifting** measures
  - → Extensive margin: firms with subsidiaries in the tax-haven, identified by Tørsløv et al., (2022)
  - → Intensive margin: share of profits shifted estimated by Delis et al. (2025)

Cross-sectional relationship

Empirics

Dependent Var	Intangible Intensity					
	(1)	(2)	(3)	(4)	(5)	(6)
MNE indicator	0.09***	0.12***	0.06***			
	(0.01)	(0.01)	(0.00)			
× post−2005	0.05***	0.05***	0.05***			
	(0.01)	(0.01)	(0.01)			
TA-region indicator				0.03***	0.04***	0.03***
				(0.01)	(0.01)	(0.01)
× post−2005				0.03***	0.04***	0.02**
				(0.01)	(0.01)	(0.01)
Industry FE	N	N	Y	N	N	Y
Control for log revenue	N	Y	Y	N	Y	Y
Observations	27,318	27,318	27,318	11,925	11,925	11,925

Note: All specifications control for year FE. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

- MNEs and profit-shifting: ↑ **intangible intensity**, also true on the intensive margin ▸ intensive
- These gaps have been increasing over time

▸ details

Cross-sectional relationship

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- MNEs have **6 p.p. higher** intangible intensity before 2005 for our preferred specification in (3)
- This gap almost doubles post-2005

details

Cross-sectional relationship

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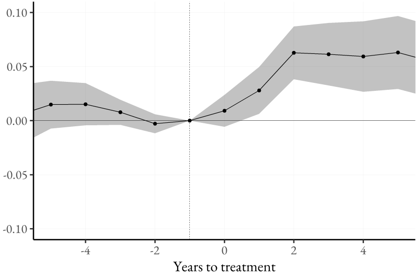
- MNEs with tax-haven subsidiaries have **3 p.p. higher** intangible intensity pre-2005 in Col. (6)
- This gap almost doubles post-2005

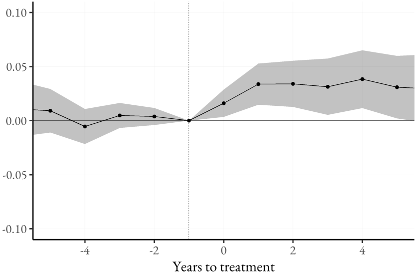
details

Event studies

Empirics



(a) MNE



(b) Profit Shifting

MNE expansion increases intangible intensity by $\approx$ **6 p.p. after 2-4 years**

▸ details ▸ add. details

Event studies

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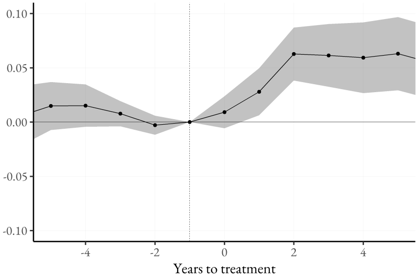
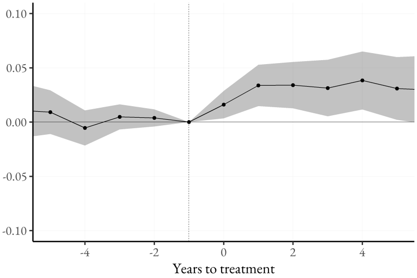


(a) MNE



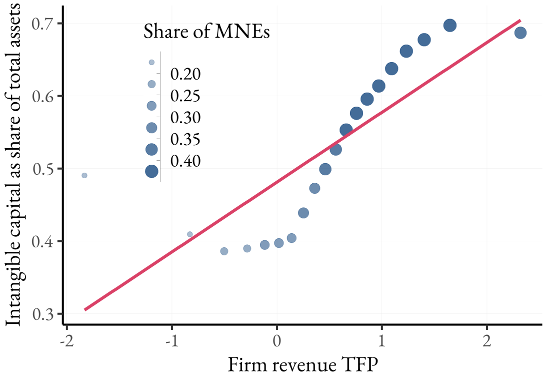
(b) Profit Shifting

MNEs engaging in profit shifting increase intangible intensity by $\approx$ **3 p.p. after 2-4 years**

details add. details

Productivity and Intangible Intensity

Empirics

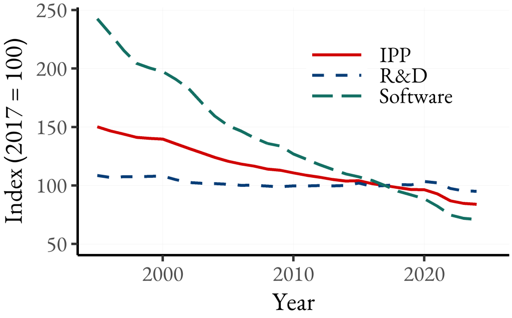


- Intangible capital share increases with firm-level productivity (TFP).
- **Evidence of selection:** MNEs tend to be more productive and intangible-heavy firms.
- Together with the event studies: selection (high-TFP firms become MNEs) and treatment (MNE entry raises intensity).

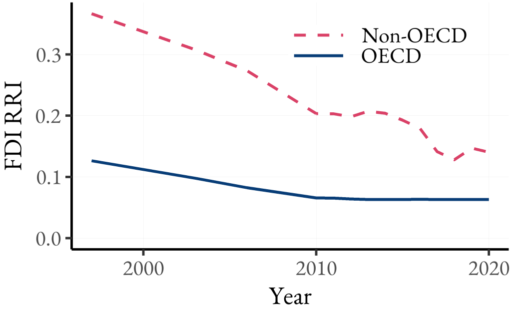
▸ ACF details

Declining intangibles costs and FDI frictions

Empirics



(a) Intangibles costs



(b) FDI restrictions

Note. BEA Price Index for Private Fixed Investment in Intellectual Property Products (IPP) and OECD FDI Regulatory Restrictiveness Index. Price indices for IPPs are deflated by CPI and normalized to 100 in 2017.

- Both intangibles costs and FDI frictions have been declining over time.

Theory

Environment

Theory

Suppose a world consisting of a set of identical, atomistic productive countries and a tax haven. Consider the following problem of a firm with productivity a:

$$\pi(a) = \max_{J,z,\lambda,l} \left\{ (1-\tau^e(\lambda)) \cdot J\pi(a,l,z) - (1-\tau)\left(MC \cdot z + C^{\lambda}(\lambda, J\pi) + C^{J}(J)\right) \right\}$$

s.t.

$$\tau^e(\lambda, s^z) = (1-\lambda s^z)\,\tau + \lambda s^z \tau_{TH}$$

$$\pi(a,l,z) = paF(z,l) - wl$$

where

- J is the measure of markets the firm operates in
- z is **nonrival** intangible capital, l is labor, $F(z,l)$ is the DRS production technology
- λ is the profit shifting intensity, τ^e is the effective tax rate ▸ DHS
- MC is the marginal cost of z, $C^{\lambda}(\cdot)$ and $C^{J}(\cdot)$ are profit shifting and expansion costs ▸ assumption

Intangible intensity and substitutability

Theory

The production function is CES in the two factors:

$$F\left(z,l\right)=\left[\mu z^{\frac{\sigma-1}{\sigma}}+\left(1-\mu\right)l^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma\alpha}{\sigma-1}}$$

We define intangible intensity as the intangible share of worldwide income:

$$s^{z}\equiv\frac{MC\cdot z}{MC\cdot z+Jwl}$$

How does it depend on model parameters?

- $\sigma=1$: always equals μ
- $\sigma>1$: changes **endogenously** in changes in firm productivity a, intangible cost MC, expansion cost C^{J}, and profit shifting cost C^{λ}

Intangibles and multinational expansion reinforce each other

Theory

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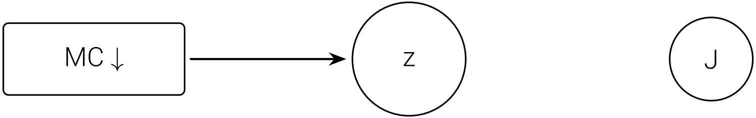

- Firms **substitute** towards intangible capital ($z \uparrow$) as the MC of intangibles decreases ($MC \downarrow$)

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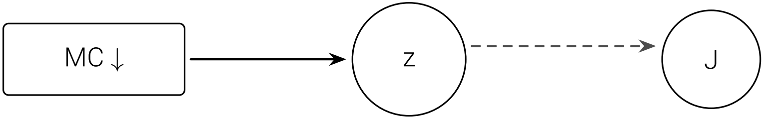

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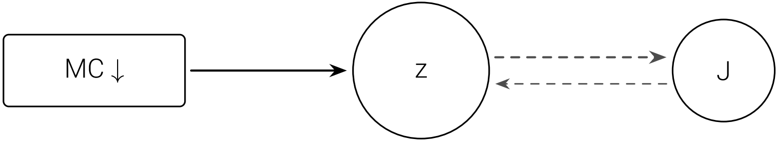

- Firms **substitute** towards intangible capital ($z \uparrow$) as the MC of intangibles decreases ($MC \downarrow$)
- The **non-rivalry** of intangible capital incentivizes multinational expansion ($z \uparrow \implies J \uparrow$).
- Greater multinational reach ($J \uparrow$) increases the return to intangible capital, **reinforcing** the impact on intangibles ($z \uparrow$).

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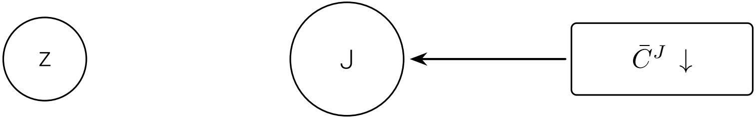

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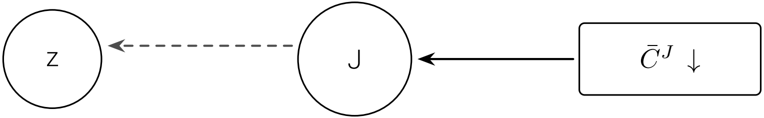

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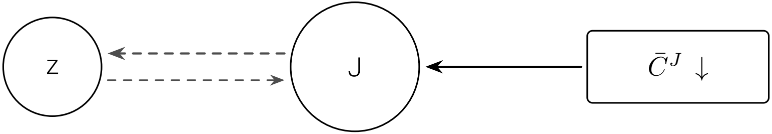

- Falling expansion costs ($\bar{C}^J \downarrow$) increases firm incentives to **expand** ($J \uparrow$)
- Greater multinational reach increases the **return to intangible capital** ($J \uparrow \implies z \uparrow$)
- A tilt towards intangibles ($z \uparrow$) makes MNE expansion even **more attractive** ($J \uparrow$)

Intangibles and multinational expansion reinforce each other

Theory

- **Concurrently** falling MC of intangible capital and expansion costs have the direct effects of tilting firms towards intangible capital and increasing multinational expansion

Intangibles and multinational expansion reinforce each other

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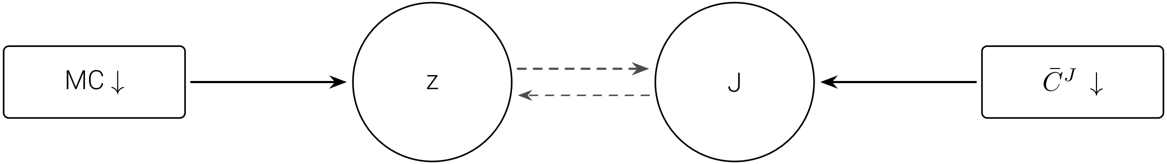
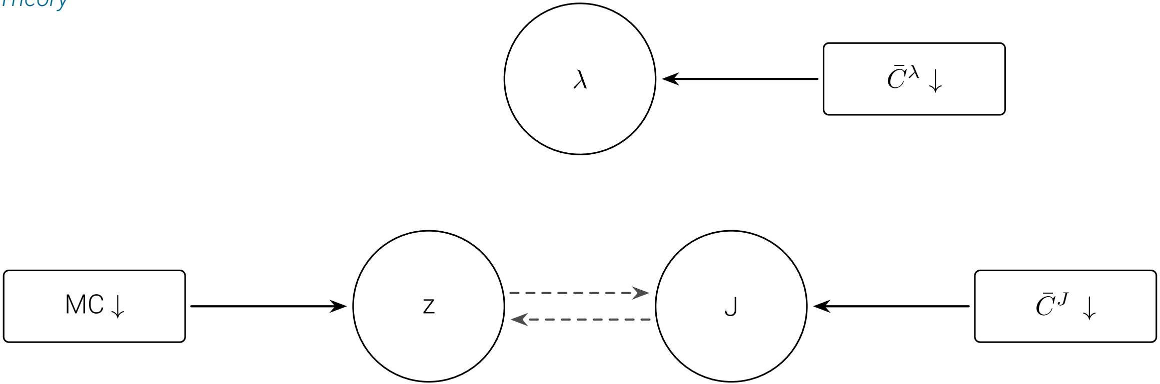


- **Concurrently** falling MC of intangible capital and expansion costs have the direct effects of tilting firms towards intangible capital and increasing multinational expansion
- Mutual reinforcement results in greater z and J than each decrease in isolation –**complementarity**.

Enter profit shifting

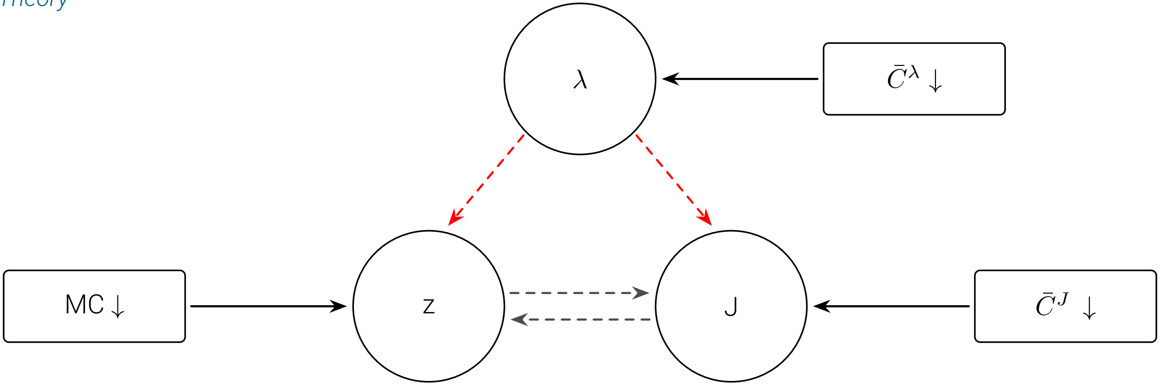
Theory



- Decreasing profit-shifting costs $\bar{C}^{\lambda}$ complements the impact of decreases in other two costs:

Enter profit shifting

Theory



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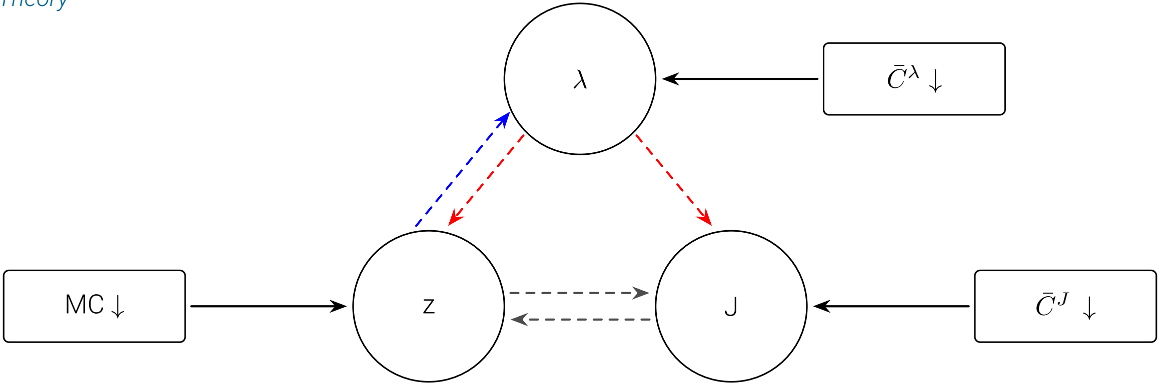
$\longrightarrow$ greater λ increases the after-tax return of intangible investment and expanding

Enter profit shifting

Theory



- Decreasing profit-shifting costs $\bar{C}^{\lambda}$ complements the impact of decreases in other two costs:
  - $\longrightarrow$ greater λ increases the after-tax return of intangible investment and expanding
  - $\longrightarrow$ greater intangible investment increases the return to profit shifting

What the theory delivers

Theory

- **Monotonicity**: Reductions in MC, $\bar{C}^J$, and $\bar{C}^\lambda$ all increase z, J, and λ. detail

- **Complementarity**: The positive effects of reductions in MC, $\bar{C}^J$, and $\bar{C}^\lambda$ on s^z, J, and λ are complementary to each other. detail

- **Firm Heterogeneity**: detail
  1. Higher productivity a firms have higher s^z, J, and λ.
  2. The positive effects of reductions in MC, $\bar{C}^J$, and $\bar{C}^\lambda$ on s^z, J and λ are stronger for high productivity firms.

Quantitative Model

What the model is designed to do

Quantitative Model

Unify and quantify the joint rise of intangibles, multinational expansion, and profit shifting:

- **Go beyond theory:**
  - $\rightarrow$ Discipline mechanisms with data and measure magnitudes.
  - $\rightarrow$ Quantitative advantage: calibrate the key forces to match firm-level and aggregate facts.
- **Decompose and experiment:**
  - $\rightarrow$ Separate effects of technological change (falling MC), globalization (lower $\bar{C}_J$, ψ), and tax policy (profit-shifting incentives).
  - $\rightarrow$ Evaluate how they reinforce each other through scalable intangibles.
- **Deliver quantitative answers:**
  - $\rightarrow$ How much of the observed trends each driver explains.
  - $\rightarrow$ How much comes from complementarity—superlinear interactions of costs.

Dynamic, heterogeneous firms model

Quantitative Model

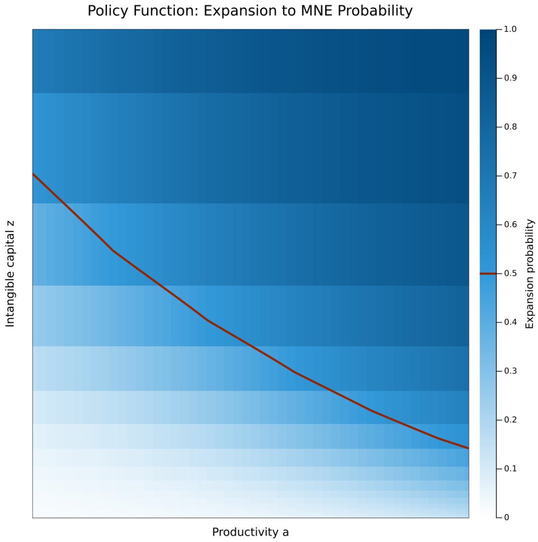
Core features:

- A firm is a triplet of individual states: (a, z, J)
  - $\rightarrow$ a: stochastic, persistent productivity
  - $\rightarrow$ z: intangible capital, nonrival within a firm
  - $\rightarrow$ J: number of foreign subsidiaries
- Choices:
  - $\rightarrow$ Intangible investment subject **marginal cost** MC and quadratic adjustment costs.
  - $\rightarrow$ Rival production factors (capital, labor).
  - $\rightarrow$ Whether to expand or contract across locations subject to **costs** $\bar{\mathbf{C}}^{\mathbf{J}}$. Expansion affected by **FDI barriers** η.
- Aggregation:
  - $\rightarrow$ Distribution of firms over individual states.
  - $\rightarrow$ Endogenous selection splits firm into two sets: non-MNES ($J = 0$) and MNEs ($J > 0$).

details

High-productivity, intangible-rich firms expand to become MNEs

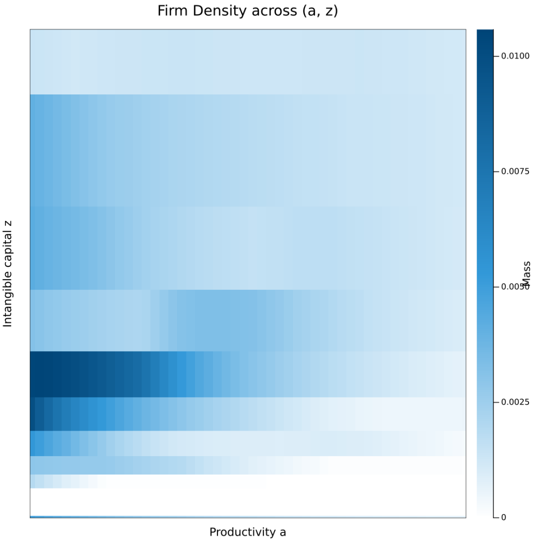
Quantitative Model

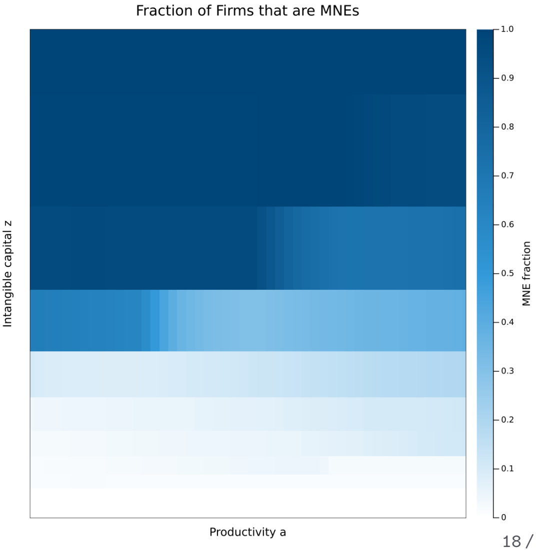


- Expansion likelihood rises with productivity and intangible capital.
- High-a, high-z firms have higher propensity to become MNEs.

Selection in the stationary distribution

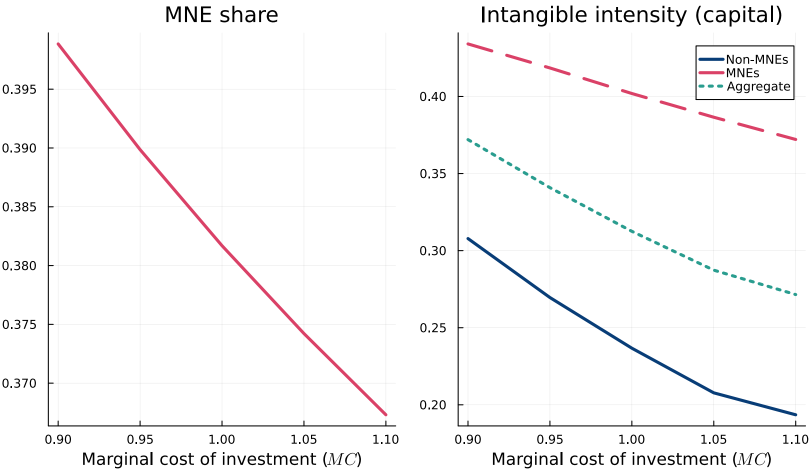
Quantitative Model





Fall in the marginal cost of intangible investment MC

Quantitative Model



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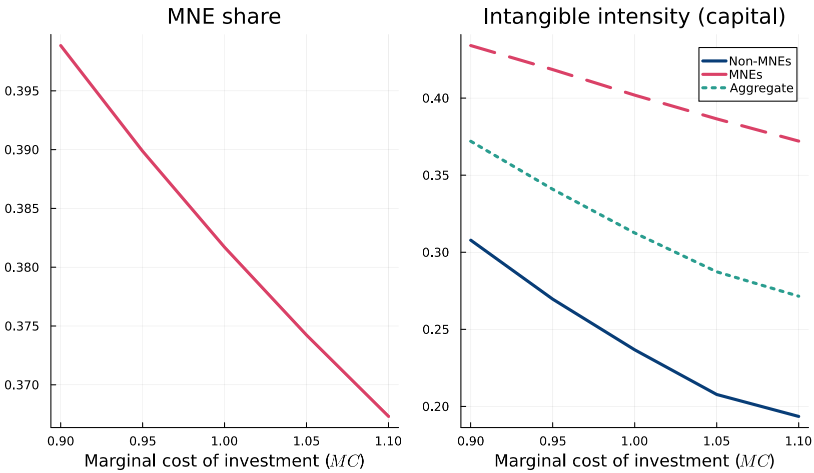
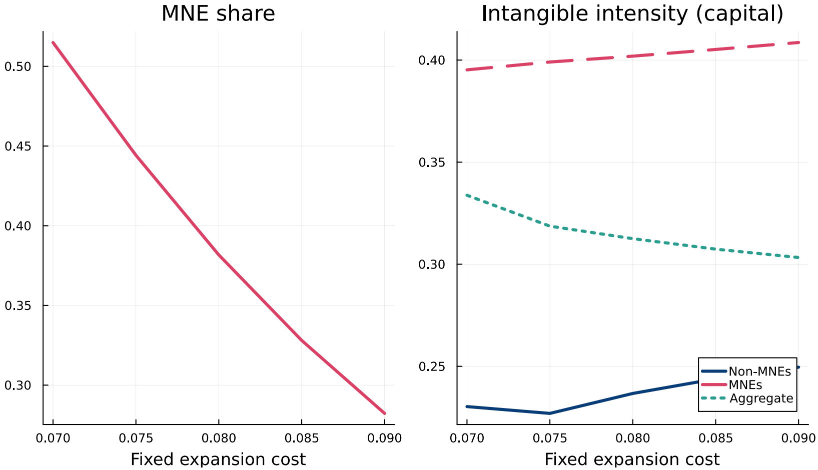


As MC falls, firms invest more in z, strengthening the z–J complementarity; expansion becomes more attractive, increasing the MNE share.

Fall in the expansion costs $\bar{C}^J$

Quantitative Model



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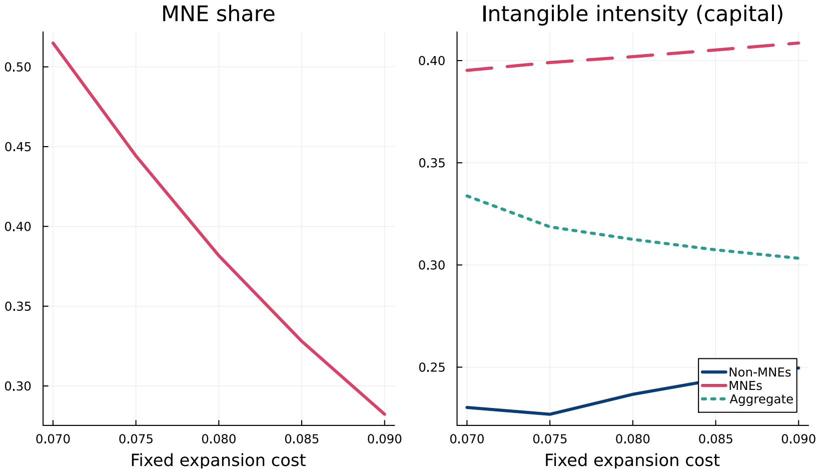
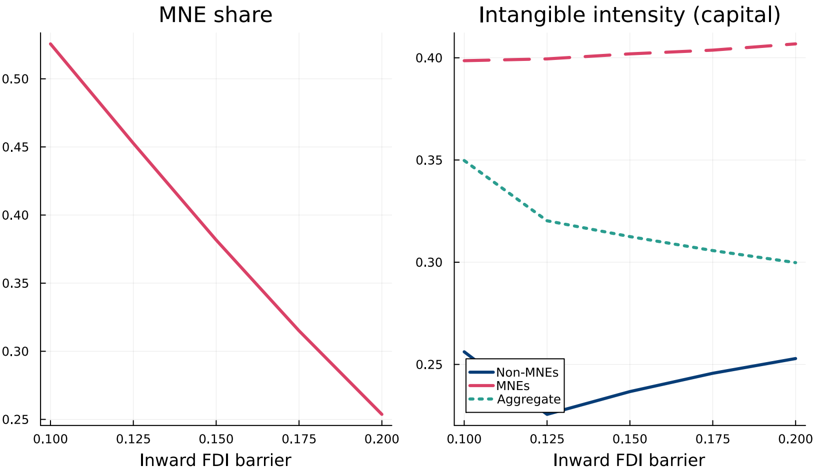


Lower expansion costs pull marginal, less productive firms into MNE status—so the MNE share and aggregate intensity rise, while the average intensity among MNEs falls.

Fall in the FDI barriers η

Quantitative Model



Fall in the FDI barriers η

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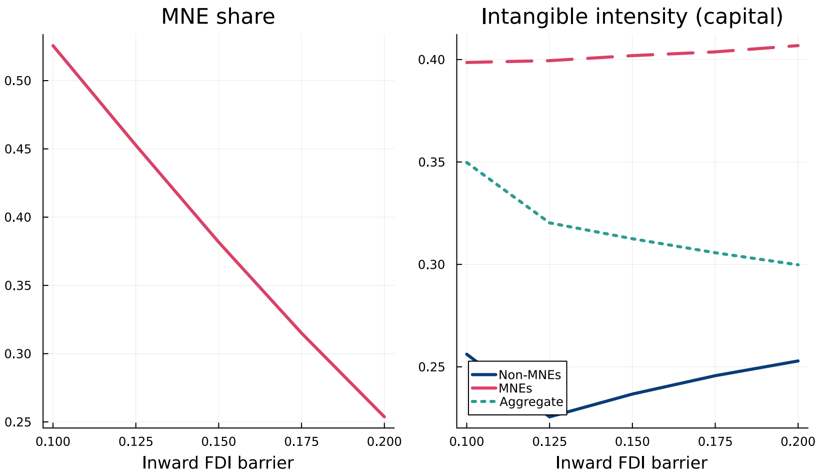

Similarly, as FDI barriers fall, more marginal firms become MNEs, boosting the MNE share and aggregate intensity while reducing average intensity among MNEs.

Testing complementarities: MC and $\bar{C}^J$

Quantitative Model

	High Investment Cost	Low Investment Cost
	% changes from High-High	
Panel (A): MNE share		
High Expansion Cost	–	$+3.0$
Low Expansion Cost	$+36.7$	$+43.1$
Complementarity (p.p.):		3.4
Panel (B): Intangible Intensity		
High Expansion Cost	–	$+12.3$
Low Expansion Cost	$+1.9$	$+20.1$
Complementarity (p.p.):		6.0

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Fall in expansion cost and drop in MC drive up MNE share and intangible intensity, and...

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... their joint effect is larger than the sum of the two (complementarity).

Key Takeaways

What to Remember

1. **Three defining trends** since the 1990s:
   - $\rightarrow$ Intangible intensity $\uparrow$
   - $\rightarrow$ Multinational reach $\uparrow$
   - $\rightarrow$ Profit-shifting $\uparrow$
2. **Empirical link**: Cross-sectional regressions and firm-level event studies confirm these moves are tightly correlated.
3. **Mechanism**: Falling R&D costs and lower FDI barriers jointly drive more intangibles and more foreign affiliates.
4. **Complementarity**: These forces reinforce each other—the combined effect exceeds the sum of their individual impacts.

Next step: use the model to run a horse race between the exogenous driving forces to decompose their effects on the endogenous trends of interest.

Appendix: Empirics

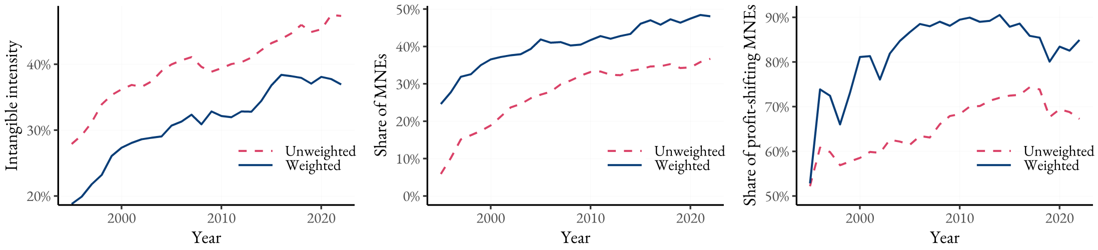
Data ▸ return

Empirics

- Firm balance-sheet data from Compustat North America
  - → Consolidated financial statement
  - → Information on sales, tangible and intangible capital, investment, foreign income, and tax liabilities
- Intangible capital
  - → Balance-sheet measure includes mostly externally-acquired intangibles
  - → Internally-generated intangible capital measured using the perpetual inventory method from R&D expenditures (Peters and Taylor, 2017)
- Multinational activity
  - → Extensive margin: firms that have foreign subsidiaries reported in Exhibit 21 of 10-K forms
  - → Intensive margin: the number of foreign countries and the share of foreign income
- Profit-shifting measures
  - → Extensive margin: firms with subsidiaries in the tax-haven, identified by Tørsløv et al., (2022)
  - → Intensive margin: share of profits shifted estimated by Delis et al. (2025)

Rise in intangibles, MNE, and profit shifting ▸ return

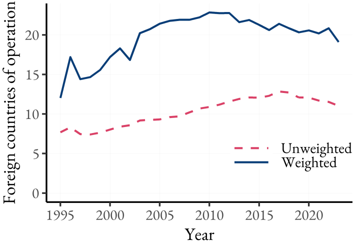
Empirics



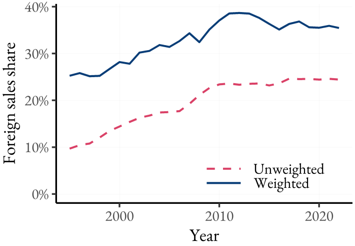
(a) Intangible Intensity   *(b)* Share of MNEs   *(c)* Share of Profit-Shifting MNEs

- Intangible Intensity = Intangible capital / (Intangible capital + Tangible capital)
  → Figure (a) is robust to (1) using intangible-revenue ratio, (2) using only knowledge capital, and (3) controlling for industry composition ▸ industry
- Intensive margin: ↑ number of foreign subsidiaries and profit-shifting intensity ▸ intensive

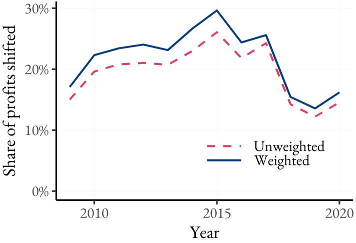
Intensive margin: multinational expansion and profit shifting ▸ return



(a) Average # of Subsidiary Countries



(b) Foreign Share of Total Income

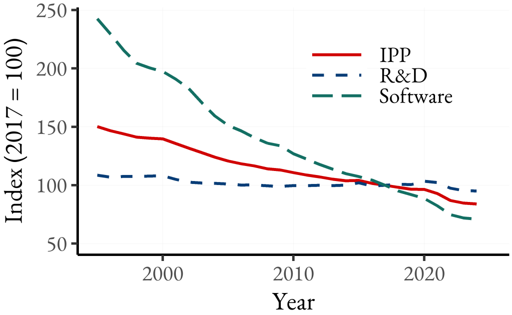


(c) Share of Profits Shifted

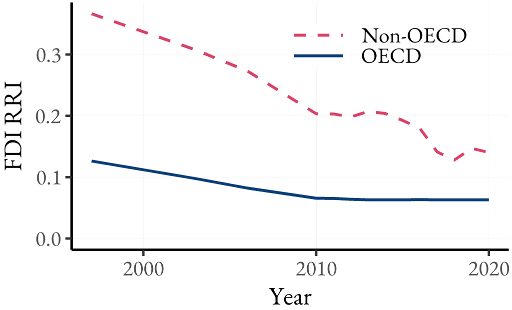
- Sub-figure (c) uses firm-year estimates of profit-shifting ratio from Delis et al. (2025).
- The steep drop in shifted profits follows the passage of the Tax Cuts and Jobs Act in 2017.

Declining intangibles costs and FDI frictions

Empirics



(a) Intangibles costs



(b) FDI restrictions

Notes: BEA Price Index for Private Fixed Investment in Intellectual Property Products (IPP) and OECD FDI Regulatory Restrictiveness Index. Price indices for IPPs are deflated by CPI and normalized to 100 in 2017.

- Both intangibles costs and FDI frictions have been declining over time.

Details on the cross-sectional regressions ▸ return

We estimate the following regression

$$\text{Intangible intensity}_{jt} = \alpha + \beta_1 \mathbb{1}_{\{Post-2005_{jt}\}} + \beta_2 \mathbb{1}_{\{X_{jt}\}} + \beta_3 \mathbb{1}_{\{Post-2005_{jt}\}} \times \mathbb{1}_{\{X_{jt}\}} + Z_{jt}\delta + \varepsilon_{jt}$$

- $\mathbb{1}_{\{X_{jt}\}}$ is an indicator for whether firm j in year t is an MNE or has a subsidiary in a tax-avoidance region
- $\mathbb{1}_{\{Post-2005_{jt}\}}$ is a post-2005 time indicator. More flexible time trend indicators yield similar results.
- Z_{jt} is a vector of controls. In the fullest specifications, we include:
  - → year FEs to account for the secular increase in intangible intensity over our analysis period,
  - → industry FEs to account for systematic variation in intangible intensity across industries (for e.g., differences in production technology, or systematic over/underreporting of intangibles on firm balance sheets from accounting conventions), and
  - → log firm revenue to proxy for time-varying firm characteristics, such as idiosyncratic productivity
- Estimates for β_2 for each X_{jt} are reported, as well as the interaction with the post-2005 indicator, β_3

Details on the event study ▸ return

We run the following event-study regression

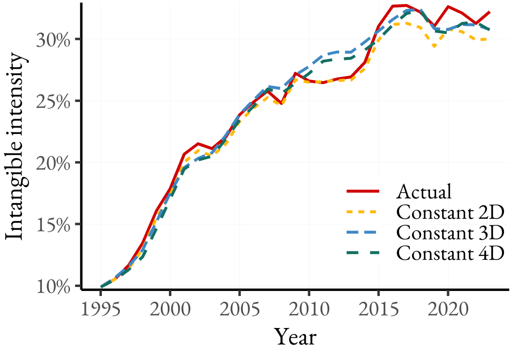
$$\text{Intangible intensity}_{jt} = \gamma_j + \delta_t + \sum_{e \notin C} \sum_{\tau \neq -1} \beta_{e,\tau} \cdot \mathbb{1}_{\{E_j = e\}} \cdot \mathbb{1}_{\{t - E_i = \tau\}} + \varepsilon_{jt},$$

- E_j is the year when firm j becomes an MNE or opens the first tax-haven subsidiary
- $\beta_{e,\tau}$ is the cohort-specific average treatment effect on the treated (CATT) τ periods from initial treatment for the cohort of firms first treated in year e
  - $\rightarrow$ The dynamic treatment effect estimate for each relative time period is given by the average of $\hat{\beta}_{e,\tau}$, weighted by the proportion of each cohort e in the relative time period τ
- γ_j and δ_t are firm and year fixed effects

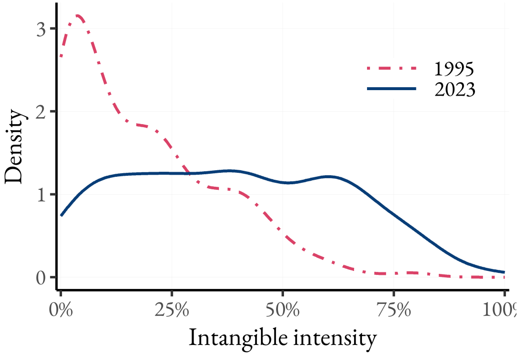
Details on the event study ▸ return

- The “building blocks" for the event study estimates are the cohort-specific average treatment effect on the treated, $\beta_{e,t}$, which are identified under a parallel trends and no anticipation assumption.
- $\beta_{e,t}$ is then averaged across all cohorts e for a given relative time, t, using weights that are the sample share of each cohort e
- The IW estimator is robust to treatment effect heterogeneity, which is potentially problematic for standard TWFE regressions.
- The parallel trends assumption seems feasible in this context since the leads of the treatment are statistically insignificant from zero.

More on Intangible Intensity with ▸ return



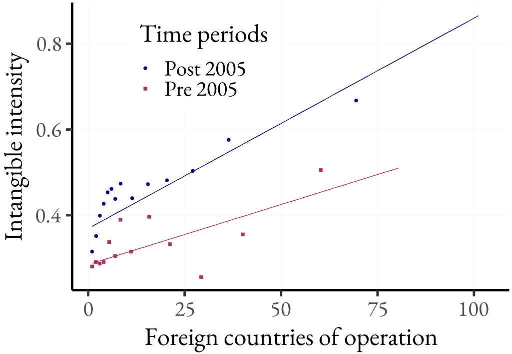
(a) Intang. intensity w. fixed ind. composition



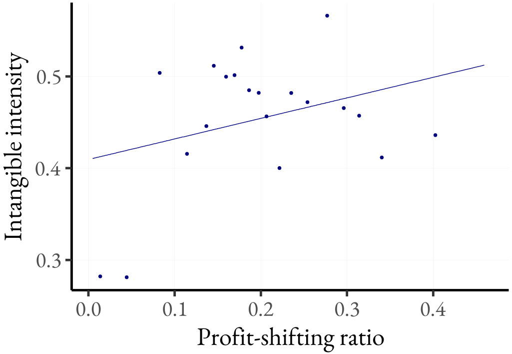
(b) Density plot

- Intangible intensity rises mainly within industries, rather than shifts towards high-intangible industries

Intensity margin: intangible intensity vs expansion and profit shifting



(a) Number of foreign countries

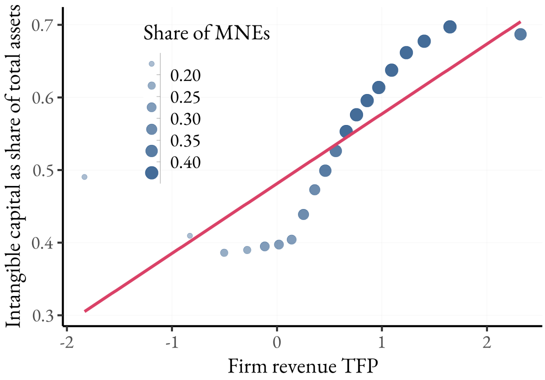


(b) Share of Shifted Profits by Delis et al. (2025)

- The firm-year level shares of shifted profits are only estimated since 2009

▸ return

Empirical Relationship Between Productivity and Intangible Intensity



- Firm-level productivity (TFP) increases with intangible capital share.
- Supports model selection: high-a, high-z firms are larger and more likely to become MNEs.

ACF details

Firm Revenue TFP estimation ▸ return

- Assume a (log) Cobb-Douglas production technology for firm i at time t

$$y_{it} = \alpha + w_{it}\beta + k_{it}\gamma + \omega_{it} + \varepsilon_{it}$$

where y_{it} is the log output; w_{it} is a $1 \times J$ vector of log free variables; k_{it} is a $1 \times K$ vector of state variables; ω_{it} is productivity; and, ε_{it} is an idiosyncratic error term.

- We use the Ackerberg et al. (2015) estimator with
  - → Value-added as output
  - → Tangible and intangible capital as the state variables
  - → Materials, approximated by `COGS`, as the proxy variable (Levinsohn and Petrin, 2003)
  - → Employment headcount, `EMP` as the free variable

to estimate firm-level productivity, $\hat{\omega}_{it}$.

Appendix: Theory

A theory of intangibles, multinational expansion and profit shifting

Theory

Suppose a world consisting of a set of identical, atomistic productive countries and a tax haven. Consider the following problem of a firm with productivity a:

$$\pi(a) = \max_{J,z,\lambda,l} \left\{ (1 - \tau^e(\lambda)) \cdot J\pi(a,l,z) - (1-\tau)\left(MC \cdot z + C^{\lambda}(\lambda, J\pi) + C^{J}(J) \right) \right\}$$

s.t.

$$\tau^e(\lambda, s^z) = (1 - \lambda s^z)\,\tau + \lambda s^z \tau_{TH}$$

$$\pi(a,l,z) = paF(z,l) - wl$$

where

- J is the measure of markets the firm operates in
- z is **nonrival** intangible capital, l is labor, $F(z,l)$ is the DRS production technology
- λ is the profit shifting intensity, effect of λ on the effective tax rate τ^e increases with intangible intensity $s^z \equiv \frac{MC \cdot z}{MC \cdot z + Jwl}$
- MC is the marginal cost of z, $C^{\lambda}(\cdot)$ and $C^{J}(\cdot)$ are profit shifting and expansion costs

Linking to Dyrda, Hong and Steinberg (2024, DHS) return

DHS establishes a micro-foundation of profit shifting using transfer pricing of intangible capital. In this simple framework,

$$\pi(a) = \max_{l,z,\lambda,J} \left\{ (1-\tau) J\hat{\pi}(a,l,z) + (1-\lambda)(1-\tau) V_i(z) + \lambda \left(1-\tau^{TH}\right) V_i(z) - Costs \right\}$$

where $\hat{\pi}(a,l,z) = p \times aF(z,l) - wl - \vartheta(z) z$, $\vartheta(z)$ is the subsidiary-level marginal product of z, $V_i(z) = J\vartheta(z)z$ is the total value of intangible capital, λ is the share of intangibles shifted to the tax haven

We can simplify $\pi(a)$ as:

$$\pi(a) = \max_{l,z,\lambda,J} \left\{ \left[1 - \underbrace{\left(\left(1 - \lambda \frac{J\vartheta(z) z}{J\pi(a,z,l)} \right) \tau + \lambda \frac{J\vartheta(z) z}{J\pi(a,z,l)} \tau^{TH} \right)}_{\text{Effective tax rate: } \tau^e} \right] J\pi(a,z,l) - Costs \right\}$$

It is straightforward to see that τ^e decreases with λ and intangible intensity.

Functional form assumptions ▸ back

Theory

We make the following parametric form assumptions on the production and cost functions:

1. The production function is CES in the two factors:

$$F(z,l) = \left[\mu z^{\frac{\sigma-1}{\sigma}} + (1-\mu)\, l^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma\alpha}{\sigma-1}}$$

   where α is the returns-to-scale, σ is the EoS between z and l, and μ is the input weight of z. We assume $\alpha < 1$, $\sigma > 1$ and the weight on rival input, $1-\mu$, is significantly large.

2. The expansion cost is iso-elastic in the number of foreign countries:

$$C^J(J) = \bar{C}^J \cdot J^\gamma$$

   with $\gamma - 1 > \frac{\partial \ln \pi}{\partial \ln J}$, i.e. the expansion cost increases faster than π as J increases.

3. The profit shifting costs is

$$C^\lambda(\lambda, J\pi) = \bar{C}^\lambda \cdot (\lambda - (1-\lambda)\log(1-\lambda)) \cdot J\pi.$$

Solutions to the key endogenous variables

Theory

With the function form assumptions, we can solve for intangible intensity s^z, number of markets J, and profit shifting intensity λ as:

$$s^z = \left[\left(\tilde{\tau} \left(\lambda, s^z \right) \frac{1-\mu}{\mu} \right)^{\sigma} \left(\frac{w}{MC} \right)^{1-\sigma} J^{1-\sigma} + 1 \right]^{-1} \tag{1}$$

$$J = \left(\frac{1}{\gamma \tilde{\tau} \left(\lambda, s^z \right) \bar{C}^J} \pi \left(a, l, z \right) \right)^{\frac{1}{\gamma - 1}} \tag{2}$$

$$\lambda = 1 - \exp \left(\frac{\left(\tau - \tau_{TH} \right) s^z}{\left(1 - \tau \right) \bar{C}^{\lambda}} \right) \tag{3}$$

Definitions

Theory

Definition

Before presenting the theoretical results, we define the elasticity terms as follows:

1. The elasticity of an endogenous variable y with respect to a variable β is defined as

$$\varepsilon_{y,\beta} = \frac{\partial \ln y}{\partial \ln \beta}.$$

2. The combined elasticity of an endogenous variable y with respect to variables β_1 and β_2 is defined as

$$\varepsilon_{y,(\beta_1,\beta_2)} = \varepsilon_{y,\beta_1} + \varepsilon_{y,\beta_2} + \frac{\partial^2 \ln y}{\partial \ln \beta_1 \partial \ln \beta_2}.$$

Theoretical result: monotonicity ▸ return

Proposition (Monotonicity)

The following results hold when the cost parameter $\beta \in \{MC, \bar{C}^J, \bar{C}^\lambda\}$ decreases:

1. *All firms increase in intangible intensity, i.e., $\epsilon_{s^z,\beta} < 0$.*
2. *All firms expand into more markets, i.e., $\epsilon_{J,\beta} < 0$.*
3. *All firms increase profit shifting intensity, i.e., $\epsilon_{\lambda,\beta} < 0$.*

Theoretical result: complementarity of cost reductions ▸ return

Proposition (Complementarity)

The following statements are true for $y = \{s^z, J, \lambda\}$ *:*

1. *The positive effects of reductions in* MC*,* $\bar{C}^J$ *are complementary to each other:*

$$\left|\varepsilon_{y,\left(\bar{C}^J, MC\right)}\right| > \left|\varepsilon_{y,\bar{C}^J}\right| + \left|\varepsilon_{y,MC}\right|.$$

2. *The positive effects of reductions in* $\bar{C}^J$*,* $\bar{C}^\lambda$ *are complementary to each other:*

$$\left|\varepsilon_{y,\left(\bar{C}^J, \bar{C}^\lambda\right)}\right| > \left|\varepsilon_{y,\bar{C}^J}\right| + \left|\varepsilon_{y,\bar{C}^\lambda}\right|.$$

3. *The positive effects of reductions in* MC*,* $\bar{C}^\lambda$ *are complementary to each other:*

$$\left|\varepsilon_{y,\left(MC, \bar{C}^\lambda\right)}\right| > \left|\varepsilon_{y,MC}\right| + \left|\varepsilon_{y,\bar{C}^\lambda}\right|.$$

Theoretical result: firm heterogeneity ▸ return

Proposition (Firm heterogeneity)

The following statements are true:

1. *Higher productivity firms have greater intangible intensity* s^z, *i.e.,* $\varepsilon_{s^z,a} > 0$.
2. *Higher productivity firms operate in more foreign markets* J, *i.e.,* $\varepsilon_{J,a} > 0$,
3. *Higher productivity firms have profit shifting intensity* λ, *i.e.,* $\varepsilon_{\lambda,a} > 0$.

Proposition (Heterogeneity in the firm-level effects of cost reduction)

The following statements are true for $y = \{s^z, J, \lambda\}$ *:*

1. *The effect of a decrease in* MC *is stronger for high-productivity firms:* $\frac{\partial^2 \log y}{\partial MC \partial a} < 0$.
2. *The effect of a decrease in* $\bar{C}^J$ *is stronger for high-productivity firms:* $\frac{\partial^2 \log y}{\partial \bar{C}^J \partial a} < 0$.
3. *The effect of a decrease in* $\bar{C}^\lambda$ *is stronger for high-productivity firms:* $\frac{\partial^2 \log y}{\partial \bar{C}^\lambda \partial a} < 0$.

Appendix: Quantification

Overview

Quantitative Model

- Finite number of symmetric high-tax countries J indexed by i, j
- Each country populated by infinitely-lived representative continuum of firms that experience idiosyncratic shocks to productivity, cost of multinational expansion, and survival
- Production technology = CES$(k^\alpha \ell^{1-\alpha},\ z)$
- Firms born with zero intangible capital and no foreign affiliates
- Fixed cost of multinational expansion drawn from Gumbel distribution to smooth out aggregates
- Intangible capital accumulation subject to firm-level adjustment cost to smooth out life-cycle dynamics
- Close model with representative consumer in each country that supplies labor inelastically and invests in tangible capital

Production technology

Quantitative Model

- Firm from country i with productivity a and intangible capital z produces output in country j:

$$f_{ij}(a, z, k_j, \ell_j) = a\mathbb{1}_{\{i \neq j\}}(1-\eta)\left[\mu z^{\frac{\sigma-1}{\sigma}} + (1-\mu)\left(k_j^{\alpha}\ell_j^{1-\alpha}\right)^{\frac{\sigma-1}{\sigma}}\right]^{\theta}$$

  - → k_j: tangible capital rented in j
  - → ℓ_j: labor hired in j
  - → α: governs labor share of value added
  - → μ governs intangible share of value added
  - → σ: elasticity of substitution between rival and nonrival factors, same as before
  - → η: variable FDI barrier as in McGrattan and Prescott (2009, 2010)
- Consistent with INSERT CITATIONS

Dynamics: Intangibles, multinational reach & productivity

Quantitative model

- Intangible investment
  - $\rightarrow$ Firm begins period with intangible capital stock z
  - $\rightarrow$ Chooses new intangible investment x, denominated in units of home country's final good
  - $\rightarrow$ Pays adjustment cost $\varphi x^2/2$
  - $\rightarrow$ Intangible capital grows to $z' = (1-\delta)z + x$
- Multinational expansion
  - $\rightarrow$ Firm begins period with $n \in \{0, \ldots, J-1\}$ affiliates
  - $\rightarrow$ Draws iid cost $\bar{C}^J$ of expanding to $n+1$ affiliates, where $\bar{C}^J = \bar{\bar{C}}^J - \epsilon_{\bar{C}^J}$ and $\epsilon \sim$ Gumbel$(0, \sigma_{\bar{C}^J})$
  - $\rightarrow$ Chooses whether to stay at n or go to $n+1$ and pay $\bar{C}^J$
- Productivity and survival
  - $\rightarrow$ Firm begins period with productivity a
  - $\rightarrow$ At end of period, dies with probability $1-\psi$
  - $\rightarrow$ Surviving firm keeps current productivity with probability ρ_a, otherwise draws new productivity from Pareto with tail parameter σ_a, CDF $G(a)$
  - $\rightarrow$ Dying firms replaced by new firms with $z = n = 0$ and a drawn from same Pareto distribution

Static problem

Quantitative model

- Given a, z, n, choose domestic factors k_i, ℓ_i, affiliate factors k_j, ℓ_j, and profit shifting λ to maximize global after-tax profits

$$\pi_i(a,z,n) = ((1-\tau(\lambda,z,n))\left\{P_i f_{ii}(a,z,k_i,\ell_i) - W_i\ell_i - \delta k_i + \sum_{j=1}^{n}[P_j f_{ij}(a,z,k_j,\ell_j) - W_j - \delta k_j]\right\}$$
$$- r_i k_i - \sum_{j=1}^{n} r_j k_j - (1-\tau)C(\lambda,z,n)$$

- Tangible capital costs not fully tax-deductable, only depreciation allowance δk
- Intangible intensity measured same as in the data:

$$s^z(a,z,n) = \frac{z}{z + k_i + \sum_{j=1}^{n} k_j}$$

Dynamic program

Quantitative model

- Given a, z, n, choose x to maximize expected present value of dividends. Given iid Gumbel structure, can write as

$$V(a,z,n) = \max_{x,\lambda}\Big\{\pi_i(a,z,n) - (1-\tau)\left[MC(x) - \varphi x^2/2\right]$$

$$+ \beta\psi\sigma_{\bar{C}^J} \log\left[\exp\left(\frac{W(a,z',n)}{\sigma_{\bar{C}^J}}\right) + \mathbb{1}_{\{n<J-1\}}\exp\left(\frac{W(a,z',n+1)-\bar{C}^J}{\sigma_{\bar{C}^J}}\right)\right]\Big\}$$

$$W(a,z,n) = \rho_a V(a,z,n) + (1-\rho_a)\int V(a',z,n)\mathrm{d}G(a')$$

- Expansion probability:

$$p(a,z,n) = \frac{\psi\mathbb{1}_{\{n<J-1\}}\exp\left(\frac{W(a,z',n+1)-\bar{C}^J}{\sigma_{\bar{C}^J}}\right)}{\exp\left(\frac{W(a,z',n+1)}{\sigma_{\bar{C}^J}}\right) + \mathbb{1}_{\{n<J-1\}}\exp\left(\frac{W(a,z',n+1)-\bar{C}^J}{\sigma_{\bar{C}^J}}\right)}$$

Households

Quantitative model

- Supply labor $\bar{L}_i$ inelastically
- Earn labor income $W_i \ell_i$, dividends D_i, and lump-sum transfer of tax revenue T_i
- Choose consumption C_i and tangible capital K_i' to maximize

$$\sum_{t=0}^{\infty} U(C_i)$$

subject to

$$C_i + K_i' - (1-\delta)K_i = W_i \bar{L} + D_i + T_i$$

Aggregation

Quantitative model

- Labor and capital markets:

$$\bar{L}_i = \int \ell_{ii}(a,z,n)\,\mathrm{d}\Psi_i(a,z,n) + \sum_{j=1}^{J-1} \int \ell_{ji}(a,z,n)\,\mathrm{d}\Psi_j(a,z,n)$$

$$K_i = \int k_{ii}(a,z,n)\,\mathrm{d}\Psi_i(a,z,n) + \sum_{j=1}^{J-1} \int k_{ji}(a,z,n)\,\mathrm{d}\Psi_j(a,z,n)$$

- Final good:

$$\begin{aligned} Y_i :&= \int y_{ii}(a,z,n)\,\mathrm{d}\Psi_i(a,z,n) + \sum_{j=1}^{J-1} \int y_{ji}(a,z,n)\mathrm{d}\Psi_j(a,z,n) \\ &= C_i + K_i' - (1-\delta)K_i + \int \left[x_i(a,z,n)/MC + \varphi x_i(a,z,n)^2/z + p_i(a,z,n)\bar{C}^J\right]\,\mathrm{d}\Psi_i(a,z,n) \end{aligned}$$

Steady-state distribution

Quantitative model

- Let $\mathcal{A}$ and $\mathcal{Z}$ denote typical subsets of the spaces of productivity and intangible capital
- Probability of survivor transiting from (a, z, n) to $(\mathcal{A}, \mathcal{Z}, n)$ as

$$Q_i(a, z, n, \mathcal{A}, \mathcal{Z}, n+1) = \left[\rho_a \mathbb{1}_{\{a \in \mathcal{A}\}} + (1-\rho_a) G(\mathcal{A})\right] \times \mathbb{1}_{\{(1-\delta)z + x_i(a,z,n) \in \mathcal{Z}\}} \times (1 - p_i(a, z, n))$$

- Probability of survivor transiting from (a, z, n) to $(\mathcal{A}, \mathcal{Z}, n+1)$ as

$$Q_i(a, z, n, \mathcal{A}, \mathcal{Z}, n+1) = \left[\rho_a \mathbb{1}_{\{a \in \mathcal{A}\}} + (1-\rho_a) G(\mathcal{A})\right] \times \mathbb{1}_{\{(1-\delta)z + x_i(a,z,n) \in \mathcal{Z}\}} \times p_i(a, z, n)$$

- Law of motion for distribution is:

$$\Psi_i'(\mathcal{A}, \mathcal{Z}, n) = \psi \int \left[Q_i(a, z, n, \mathcal{A}, \mathcal{Z}, n) + \mathbb{1}_{\{0 \in \mathcal{Z} \wedge n=0\}} G(\mathcal{A})\right] \mathrm{d}\Psi_i(a, z, n), \; n \in \{0, \dots, J-1\}$$

$$\Psi_i'(\mathcal{A}, \mathcal{Z}, n+1) = \psi \int Q_i(a, z, n, \mathcal{A}, \mathcal{Z}, n+1) \mathrm{d}\Psi_i(a, z, n), \; n \in \{0, \dots, J-2\}$$

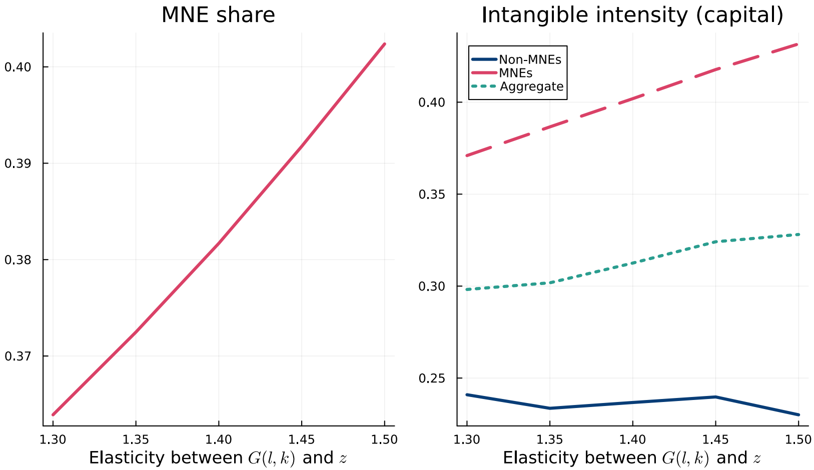
Calibration strategy

Quantitative model

- Calibrate to match post-2005 Compustat moments
- Parameters:
  - $\rightarrow$ Productivity distribution
  - $\rightarrow$ Returns to scale
  - $\rightarrow$ Elasticity of substitution between intangibles and rival factors
  - $\rightarrow$ Fixed and variable expansion costs, fixed cost shock distribution
  - $\rightarrow$ Fixed and variable profit shifting costs, fixed cost shock distribution
  - $\rightarrow$ Intangible investment adjustment cost
- Moments:
  - $\rightarrow$ Extensive margin: Fraction of firms that are non-MNEs, MNEs, profit shifters (PSE)
  - $\rightarrow$ Size distributions: Non-MNEs, MNEs, PSEs. Overlaps pin down fixed cost shock dispersion.
  - $\rightarrow$ Intangible intensity: Non-MNEs, MNEs, PSEs
  - $\rightarrow$ Event-study responses: non-MNE $\rightarrow$ MNE, non-PSE $\rightarrow$ PSE

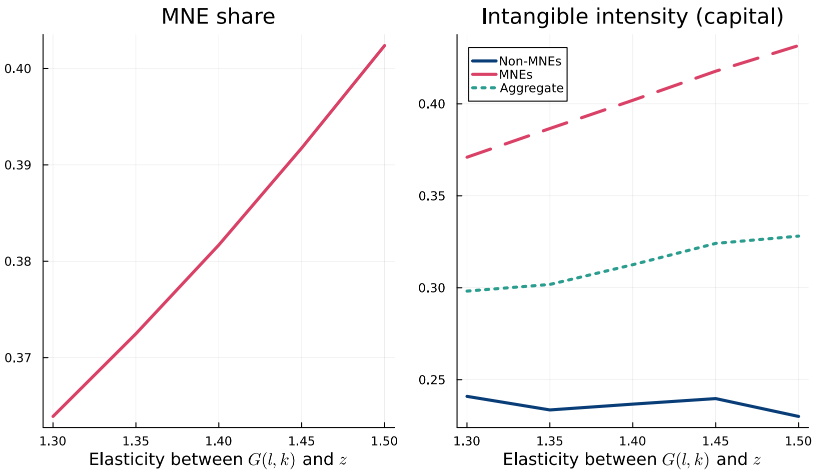
Identification of the elasticity of substitution σ_z

Quantitative Model



Identification of the elasticity of substitution σ_z

Quantitative Model



A higher σ_z makes firms substitute more strongly toward non-rival intangibles. This raises intangible intensity—especially for MNEs—and increases the MNE share.